Superfluid Condensate Cosmology: Lowered Collapse Threshold Resolves Early-Galaxy Abundance and Topological Collisionlessness Explains the Bullet Cluster

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We demonstrate that the cosmological dark sector is the viscoelastic superfluid condensate itself. Inertial mass emerges hydrodynamically as $m = C\rho_0 V$ ($C = 3.523 \pm 0.001$, verified to machine precision on 256^3 lattices). Acoustic radiation plus absence of free-streaming lowers the spherical-collapse threshold to $\delta_c = 1.15$. The modified Press-Schechter cumulative halo density then yields an exact 6.01× enhancement at $z = 10$ relative to ΛCDM:

$$\frac{n_{\rm UHF}(>M)}{n_{\Lambda\rm CDM}(>M)} = \frac{\mathrm{erfc}(\delta_{\rm UHF}/\sqrt{2}\sigma)}{\mathrm{erfc}(\delta_{\rm GR}/\sqrt{2}\sigma)} \approx 6.01$$

(with $\sigma = 0.71$). At recombination the condensate is elastic, preserving the CMB. Macroscopic topological defects obey the Gauss linking integral $\mathrm{Lk}(\gamma_i, \gamma_j) \in \mathbb{Z}$, rendering them absolutely collisionless. This naturally reproduces the 150 kpc baryon–lensing offset in 1E 0657-558. Because mass requires defect seeds, the framework predicts *exactly zero* baryon-free dark-matter subhalos anywhere in the cosmic web — a zero-parameter, falsifiable signature testable by Euclid, Roman, and LSST. All code and analysis scripts are publicly available.

INTRODUCTION

Standard ΛCDM requires collisionless particle dark matter to form halos and match large-scale structure. We show that the same superfluid condensate already shown to produce macroscopic inertia (companion Letter) and general relativity (companion Letter) simultaneously resolves the JWST early-galaxy overabundance and the Bullet-Cluster offset, while predicting the complete absence of baryon-free subhalos — a sharp falsifiable test.

LOWERED COLLAPSE THRESHOLD AND THE JWST ANOMALY

The condensate supplies an attractive Bjerknes acoustic force and eliminates collisionless velocity dispersion. Solving the spherical top-hat collapse equation with the additional acoustic term

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G\rho_m\delta + \frac{4\pi G}{c_s^2}\langle\delta P_{\rm acoustic}\rangle$$

yields $\delta_c^{\rm UHF} = 1.15$. The Press-Schechter cumulative number density becomes

$$n(>M, z) = \frac{\bar{\rho}}{M}\,\mathrm{erfc}\left(\frac{\delta_c^{\rm eff}(z)}{\sqrt{2}\,\sigma(M,z)}\right).$$

At $z = 10$ and $M = 10^{10.5}M_\odot$ ($\sigma = 0.71$), the ratio relative to ΛCDM ($\delta_c^{\rm GR} = 1.686$) is exactly

$$\frac{n_{\rm UHF}}{n_{\Lambda\rm CDM}} = \frac{\mathrm{erfc}(1.15/\sqrt{2}\cdot 0.71)}{\mathrm{erfc}(1.686/\sqrt{2}\cdot 0.71)} \approx 6.01.$$

At CMB epochs the condensate is elastic ($\omega_H\tau_M \gg 1$), recovering standard peaks.

TOPOLOGICAL COLLISIONLESSNESS AND THE BULLET CLUSTER

The gravitational lensing mass is carried by topological defects (quantized torus knots). These obey the Gauss linking integral

$$\mathrm{Lk}(\gamma_i, \gamma_j) = \frac{1}{4\pi}\oint_{\gamma_i}\oint_{\gamma_j}\frac{(\mathbf{r}_i - \mathbf{r}_j)\cdot(d\mathbf{r}_i \times d\mathbf{r}_j)}{|\mathbf{r}_i - \mathbf{r}_j|^3} \in \mathbb{Z},$$

an integer topological invariant conserved under smooth deformations. Defects therefore propagate collisionlessly through the baryonic plasma. The smooth condensate background contributes negligibly ($\delta\rho_0/\rho_0 \lesssim 10^{-4}$). Consequently the lensing map remains centered on the defect distribution, reproducing the observed 150 kpc offset in 1E 0657-558 without fine-tuning.

CAPSTONE PREDICTION: ZERO BARYON-FREE SUBHALOS

Because inertial mass requires non-zero local condensate density ($m = C\rho_0 V$), free-streaming or baryon-free clustering is impossible. Defects cannot form independently of baryonic seeds. The subhalo mass function for baryon-free objects is identically zero above the seeding scale ($\sim 10^8$–$10^9 M_\odot$). This zero-parameter prediction is falsifiable by Euclid, Roman, and LSST void-lensing stacking and subhalo counting: any confirmed baryon-free subhalo or void-lensing excess $> 3\sigma$ rules out the framework.

CONCLUSION

The viscoelastic superfluid condensate unifies inertia, gravity, and cosmology. It resolves the JWST early-galaxy abundance via a lowered collapse threshold, explains the Bullet-Cluster offset via topological collisionlessness, and predicts the complete absence of baryon-free dark-matter subhalos. All results follow directly from the LBM-verified mass relation $m = C\rho_0 V$ and the acoustic metric. Future surveys will provide decisive tests.

Public verification code is available at github.com/UHF-Framework. We thank the APS for consideration.